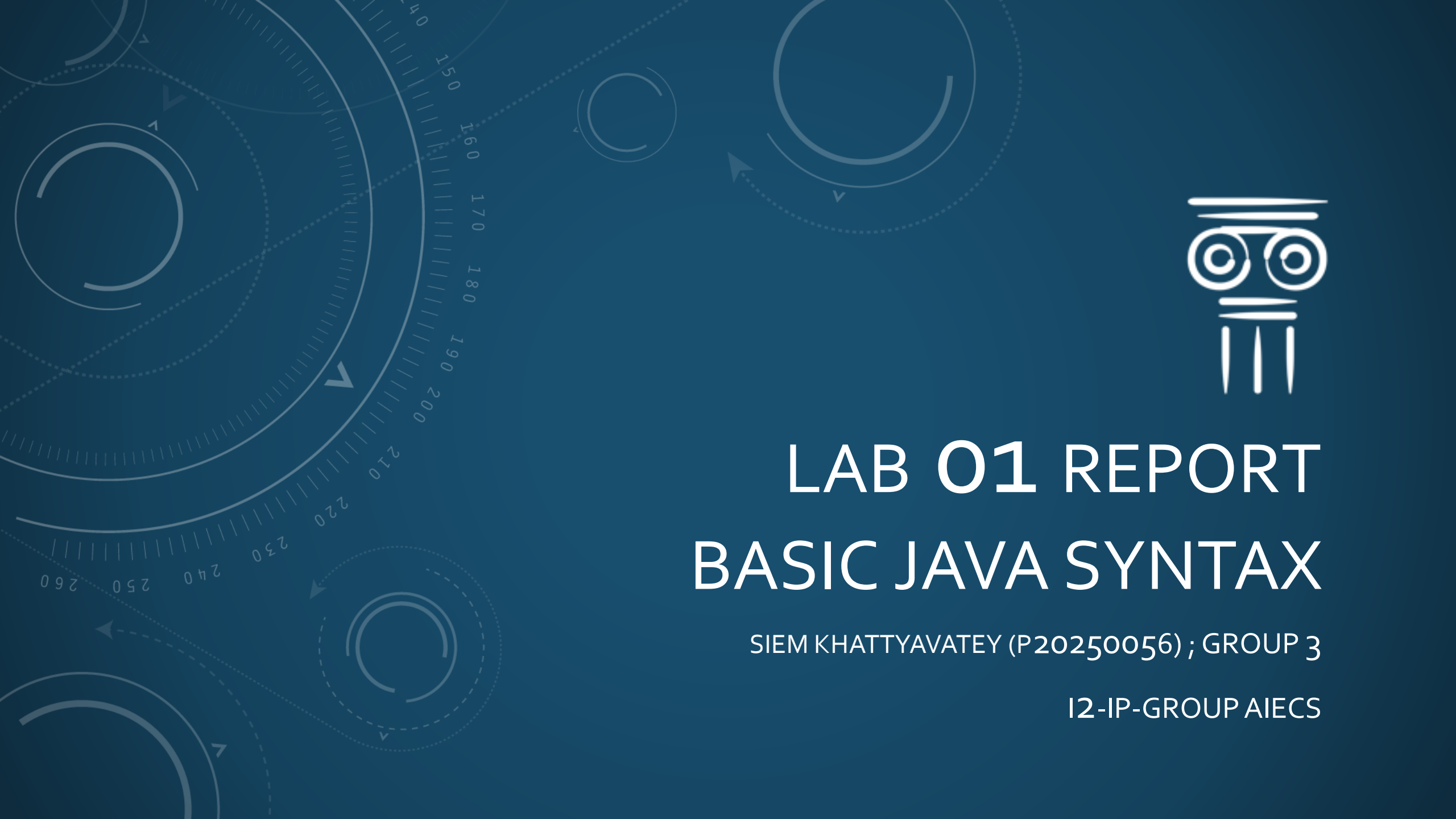


LAB 01 REPORT
BASIC JAVA SYNTAX

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```java
package test;
import java.util.Scanner;
class inputExample {
        public void main(String[] args) {
            Scanner sc=new Scanner(System.in);
            int n;
            double sum = 0;
            double average =0;
            System.out.println("Input n:");
            n=sc.nextInt();
            for(int i=1000;i>=n;i--){
                sum = sum + i;
            }
                average= sum/(1001-n);
            System.out.println("The sum is:"+sum);
            System.out.println("The average is: "+average);

        }
    }
```



EXERCISE 1

Ask a user for a number, say n. Display all sequence numbers from 1000 to n. Find the sum and the average of these numbers and display on screen.

.

OUTPUT:

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaa
Input n:
10
1000 999 998 997 996 995 994 993 992 991 990 989 98
The sum is:500455.0
The average is: 505.0

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaa
Input n:
999
1000 999
The sum is:1999.0
The average is: 999.5

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaa
Input n:
-3
1000 999 998 997 996 995 994 993 992 991 990 989 98
The sum is:500494.0
The average is: 498.5

Process finished with exit code 0
```

EXERCISE 2



Write a program to solve quadratic equation $ax^2 + bx + c = 0$ using the following formula.

Ask the input a , b , c from the user . Then the program find the roots and display it on screen

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

```java
package test;

import java.util.Scanner;

public class FindRoots {
    public void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int a;
        int b;
        int c;
        double X1=0;
        double X2=0;
        double delta=0;
        System.out.println("Input a:");
        a=sc.nextInt();
        if(a==0){
            System.out.println("a can not be 0");
        }
        else {
            System.out.println("Input b:");
            b = sc.nextInt();
            System.out.println("Input c:");
            c = sc.nextInt();
            System.out.println("The equation is " + a + "x² +" + b + "x " + c + " = 0");
            delta = ((b * b) - (4 * a * c));
            if (delta > 0)
            {
                X1 = ((-(b) + Math.sqrt(delta)) / 2 * a);
                X2 = ((-(b) + Math.sqrt(delta)) / 2 * a);
                System.out.println("The first root is " + X1);
                System.out.println("The second root is " + X2);
            }
        else if(delta == 0){
                X1 = (-(b) / 2 * a);
                System.out.println("x=" + X1);
            } else {
                System.out.println("There is no root");
            }
        }
    }
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input a:
10
Input b:
20
Input c:
1
The equation is 10x² +20x 1 = 0
The first root is -5.131670194948619
The second root is -5.131670194948619

Process finished with exit code 0
```

Delta is bigger than zero

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input a:
1
Input b:
2
Input c:
1
The equation is 1x² +2x 1 = 0
x=-1.0

Process finished with exit code 0
```

Delta is equal to zero

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input a:
5
Input b:
3
Input c:
9
The equation is 5x² +3x 9 = 0
There is no root

Process finished with exit code 0
```

Delta is smaller than zero

```java
package test;

import java.util.Scanner;

public class CompareNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Input: ");
        int a;
        int b;
        int c;
        int d;
        int e;
        int f;
        int g;
        a = sc.nextInt();
        b = sc.nextInt();
        c = sc.nextInt();
        d = sc.nextInt();
        e = sc.nextInt();
        f = a ;
        g = a;
        if(b>f){f=b;}
        if(c>f){f=c;}
        if(d>f){f=d;}
        if(e>f){f=e;}
        if(b<g){g=b;}
        if(c<g){g=c;}
        if(d<g){g=d;}
        if(e<g){g=e;}
            System.out.println("The largest number is " + f);
            System.out.println("The smallest number is " + g);
    }
}
```



EXERCISE 3

Write a program that asks the user to input 5 numbers.

The program processes the input from the user and find out the minimum and maximum numbers.

OUTPUT:

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input:
50
30
20
18
5
The largest number is 50
The smallest number is 5

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input:
5049
-74
45
72
09
The largest number is 5049
The smallest number is -74

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input:
23
53
34
23
12
The largest number is 53
The smallest number is 12

Process finished with exit code 0
```



EXERCISE 4

Write a program that can check if the given input date (yy-mm-dd) is valid or invalid.

```java
package test;

import java.util.Scanner;

class CheckDate {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int year;
        int month;
        int date;
        System.out.println("Input Year: ");
        year = sc.nextInt();
        System.out.println("Input Month: ");
        month = sc.nextInt();
        System.out.println("Input Day: ");
        date = sc.nextInt();
        System.out.println("Your input date: " + year + "-" + month + "-" + date);
            switch (month) {
                case 1, 3, 5, 7, 8, 10, 12:
                    if(date<1 || date>31){System.out.println("Invalid Date");}
                    else{System.out.println("Valid Date");}
                    break;
                case 2:
                    if((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)){
                        if(date<1 || date>29){System.out.println("Invalid Date");
                        }else{System.out.println("Valid Date");}}
                    else{
                        if(date<1 || date>28){System.out.println("Invalid Date");
                        }else{System.out.println("Valid Date");}}
                    break;
                case 4, 6, 9, 11:
                    if(date<1 || date>30){System.out.println("Invalid Date");}
                    else{System.out.println("Valid Date");}
                    break;
                default:
                System.out.println("Invalid Date");
                break;
            }
        }
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input Year:
2024
Input Month:
2
Input Day:
29
Your input date: 2024-2-29
Valid Date

Process finished with exit code 0
```

Leap year

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input Year:
2025
Input Month:
2
Input Day:
29
Your input date: 2025-2-29
Invalid Date

Process finished with exit code 0
```

Not leap year

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-javaagent:
Input Year:
2026
Input Month:
3
Input Day:
32
Your input date: 2026-3-32
Invalid Date

Process finished with exit code 0
```

Day exceed 31 day

```java
package test;

import java.util.Scanner;

class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        System.out.print("Input number: ");
        n = sc.nextInt();
        System.out.print("Prime number between 2 and " + n + " : ");
        if (n < 2) {
            System.out.println("There is none.");}
        if(n>=2) {
            for (int i = 2; i < n; i++) {
                int k = 0;
                for (int j = 2; j < i; j++) {
                    if (i % j == 0) {
                        k++;
                    }
                }
                if (k == 0) {
                    System.out.print(i + " ");
                }
            }
        }
    }
}
```



EXERCISE 5

Write a program that can display all prime number from 2 to n , where n is an input by user.

OUTPUT:

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-
Input number: 15
Prime number between 2 and 15 : 2 3 5 7 11 13
Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-
Input number: -7
Prime number between 2 and -7 : There is none.

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-
Input number: 10
Prime number between 2 and 10 : 2 3 5 7
Process finished with exit code 0
```

EXERCISE 6



Write a program that prompt the user for inputting the score of 4 subjects (Math , English, Physic, Programming).

a. Which course did the student got max score; min score?

b. What are the average score of this student?

```java
package test;

import java.util.Scanner;

class InputScore {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int math;
        int english;
        int physic;
        int program;
        double avg;
        System.out.println("Input your score(Math; English; Physic and Programing): ");
        math = sc.nextInt();
        english = sc.nextInt();
        physic = sc.nextInt();
        program = sc.nextInt();
        avg = (math + physic + english + program)/4;
        int[] numbers = {math,english,physic,program};
        String[] subjects = {"Math", "English", "Physic", "Programming"};
        int max = numbers[0];
        int min = numbers[0];
        for (int i = 1; i < numbers.length; i++) {
            if (numbers[i] > max) {
                max = numbers[i];
            }
            if (numbers[i] < min) {
                min = numbers[i];
            }
        }
        System.out.print("Highest score: " + max);
        for (int i = 0; i < numbers.length; i++) {
            if (numbers[i] == max) {
                System.out.print("("+ subjects[i]+ ") ");
            }
        }
        System.out.println();
        System.out.print("Lowest score: " + min);
        for (int i = 0; i < numbers.length; i++) {
            if (numbers[i] == min) {
                System.out.print("("+ subjects[i]+ ") ");
            }}
        System.out.println("");
        System.out.println("The average score is: " + avg);
    }
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-
Input your score(Math; English; Physic and Pro
50
40
15
100
Highest score: 100(Programming)
Lowest score: 15(Physic)
The average score is: 51.0

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-
Input your score(Math; English; Physic and Pro
40
50
60
60
Highest score: 60(Physic) (Programming)
Lowest score: 40(Math)
The average score is: 52.0

Process finished with exit code 0
```

```
"C:\Program Files\Java\jdk-27\bin\java.exe" "-
Input your score(Math; English; Physic and Pro
40
40
50
50
Highest score: 50(Physic) (Programming)
Lowest score: 40(Math) (English)
The average score is: 45.0

Process finished with exit code 0
```